

Caleb Broodo

Ph.D. Candidate, Physics Department, University of Houston (UH)

Houston TX 77204 | cbbroodo@uh.edu | [linkedin.com/in/caleb-broodo](https://www.linkedin.com/in/caleb-broodo) | [Google Scholar](#) | updated Sep. 2026

Personal Statement

I am a Ph.D. candidate in experimental heavy-ion physics at the University of Houston and a graduate student member of the STAR Collaboration at Brookhaven National Laboratory. My research focuses on extracting thermodynamic properties of quark-gluon plasma, including the speed of sound, from relativistic heavy-ion collision data. I work on the analysis and hydrodynamic interpretation of Au+Au collisions across multiple beam energies. My background combines nuclear physics data analysis with electrical engineering, electronics, and simulation, including prior work in gravitational-wave signal processing and applied electromagnetics. I played for University of Houston Men's Basketball Team that advanced to the 2021 Final Four.

Education

University of Houston, Houston, TX, 2022 – Present (Expected graduation: 2027)

Ph.D. Candidate in Physics, Experimental High Energy Nuclear Physics

Extracting the speed of sound from mean transverse momentum measurements in ultra-central Au+Au collisions with RHIC Beam Energy Scan-II program, advisor René Bellwied

M.S. in Physics (2022-2026)

University of Houston, Houston, TX, 2017 – 2022

B.S. in Electrical Engineering

Concentration in Electronics

Magna Cum Laude

Senior Design: Personal temperature regulation vest powered by Peltier effect, advisor Steven Pei

Research Experience

STAR Collaboration Graduate Student Member, Brookhaven National Laboratory, Upton, NY,
Aug. 2022 – Present

DOE SCGSR Fellow, Brookhaven National Laboratory, Upton, NY, June 2024 – Sep. 2024 & Nov.
2024 – June 2025

Extracting and interpreting the speed of sound from mean transverse momentum measurements in ultra-central Au+Au collisions with RHIC Beam Energy Scan II program, advisors René Bellwied, Lijuan Ruan (STAR spokesperson)

DOE IRCA Fellow, CERN, Geneva, CH, Sep. 2024 - Nov. 2024

Extracting and interpreting the speed of sound from mean transverse momentum measurements ultra-central Au+Au collisions using Trajectum hydrodynamical framework, advisors Alexander Kalweit (ALICE), Govert Nijs (CERN Department of Theoretical Physics)

Summer Undergraduate Research Fellow and Provost Undergraduate Research Scholarship,
University of Houston, Houston, TX, Feb. 2021 – Apr. 2022

Applying functional analysis techniques to LIGO matched-filtering method, advisors Gemunu Gunaratne and René Bellwied

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Undergraduate Electromagnetics Research, Department of Electrical and Computer Engineering, University of Houston, Houston, TX, Sept. 2021 – May 2022

Experimental and simulation study on the safety of heating due to current and current density in blood clot retriever devices provided by healthcare technology companies, advisor Ji Chen

Technical Skills

Programming and Data Analysis: C++, Python, ROOT, LaTeX, Bash, Git

Simulation and Modeling: Nuclear Physics (Trajectum, MUSIC Hydro, SMASH, TRENTTo) and Applied Electromagnetics (Ansys Maxwell, Ansys HFSS)

Electronics: Circuits, instrumentation, device-level measurement, experimental prototyping

Publications

As a member of the STAR Scientific Collaboration since 2022, I am a co-author on over 20 full-collaboration papers. Highlighted below are those which I have made a direct personal contribution:

* indicates student advisee

1. **The STAR Collaboration**, *Extracting the Energy Dependence of the Speed of Sound from Mean Transverse Momentum Measurements in Ultra-Central Au+Au Collisions at RHIC*. (Manuscript in preparation, 2026)
2. **The STAR Collaboration**, *Imaging shapes of atomic nuclei in high-energy nuclear collisions*. Nature **635**, 67-72 [<https://doi.org/10.1038/s41586-024-08097-2>] (2024)
3. **The STAR Collaboration**, “Non-Monotonicity of Transverse Momentum Correlations in Au + Au Collisions at RHIC. Phys. Rev. Lett. [<https://doi-org.ezproxy.lib.uh.edu/10.1103/2xsn-rgx3>] (2026)
4. C. Schmidt*, **C. Broodo**, R. Manikandhan, R. Bellwied, *Disentangling centrality-estimator effects in extracting the speed of sound from mean transverse momentum measurements in relativistic hydrodynamics* (Manuscript in preparation for PRC, 2026)
5. J. Parker*, **C. Broodo**, R. Manikandhan, R. Bellwied, *Null-Hypothesis testing for energy dependent Non-Monotonicity of Transverse Momentum Correlations in relativistic hydrodynamics* (Manuscript in preparation for PRC, 2026)
6. T. Vu, **C. Broodo**, V. Kashyap, A. Nageswaran, E. Skillrud, J. Chen, *Simulation of Induced Current Distribution inside a Virtual Human Model during Stroke Treatment*. IEEE (2022) [<https://doi.org/10.1109/WMCS55582.2022.9866317>]

Selected Presentations

1. **The 18th Workshop on Particle Correlations and Femtoscopy**, Eötvös Loránd University (face-to-face), Budapest, May 2026
Extracting the speed of sound in mean transverse momentum measurements in Au+Au collisions from RHIC Beam Energy Scan-II at STAR [<https://indico.cern.ch/event/1679596/contributions/7078101/>]
Proceedings (Available Oct. 2026)

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2. **The 22nd International Conference on Strangeness in Quark Matter**, University of California Los Angeles, CA (face-to-face), Mar. 2026

Extracting the speed of sound in mean transverse momentum measurements in Au+Au collisions from RHIC Beam Energy Scan-II at STAR [<https://indico.global/event/13943/contributions/144991/>]
Proceedings [<https://doi.org/10.1016/j.jspc.2026.100514>] (2026)

3. **STAR spring collaboration meeting**, Brookhaven National Laboratory, NY (face-to-face), Mar. 2026

Extracting the speed of sound in mean transverse momentum measurements in Au+Au collisions from RHIC Beam Energy Scan-II at STAR

4. **University of Houston Undergraduate Research Day**, University of Houston, Houston, TX (face-to-face), Apr. 2022

Almost-Ideal Filtering: A Demonstration of New Signal Processing Tools in GW150914 and other High Energy Compact Binary Coalescences [<https://hdl.handle.net/10657/11665>]

Intercollegiate Athletics

Division I Men's Basketball Team, University of Houston, Houston, TX, Sep. 2017 – July

2021 NCAA tournament: 2nd round (2017-18), Sweet 16 (2018-19), **Final Four** (2021)
Conference champions (2018-19), (2019-20), (2021)

Awards and Honors

DOE Office of Science Graduate Student Research Fellowship (2024 - 2025)

DOE Office of Science International Research Collaboration (2024)

UH NSM Scholarship Program (2024)

Phillips 66 Fueling Future Engineers Award (2022)

For outstanding academic accomplishments and leadership in the UH engineering community

Making a Difference Award: UT-Dallas Excellence in Senior Design Capstone Showcase (2022)

In recognition of a senior design project addressing a quality-of-life challenge through applied engineering design

Summer Undergraduate Research Fellowship (2021)

Provost's Undergraduate Research Scholarship (2021)

Certificate of Congressional Recognition from Rep. Sheila Jackson Lee, 18th District of Texas (2021)

Recognized for contributions to UH Men's Basketball 2020–2021 season – American Athletic Conference Champions, NCAA Final Four

Dean's List: (Fall 2019, Spring 2020, Fall 2020, Spring 2021, Fall 2021)

Teaching, Leadership, and Mentorship

Experimental Shift Work

Shift Leader, STAR Collaboration data-taking during run 25 (Sep. 2024)

Detector Operator, STAR Collaboration data-taking during run 24 (Jun. 2024)

Detector Operator, STAR Collaboration data-taking during run 23 (Jun. 2023)

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Research Mentorship, *Trajectum* computational collision framework at the University of Houston

Christian Schmidt, Undergraduate student, (Mar. 2025 – present)

Justin Parker, Undergraduate student, (Jul. 2025 – present)

Yahia Abuali, Undergraduate student (Jul. 2025)

Mustafa Murtaza, Undergraduate student (May 2025 – Aug. 2025)

Alex Zou, High school student (May 2025 – Aug. 2025)

Teaching Assistant, University of Houston, Aug. 2022 – Aug. 2023

Electronic Devices and Their Applications

University Physics Laboratory II

Particle Physics Journal Club, University of Houston, Jan. 2026 – May 2026

Presented *The Hyperon Puzzle with Neutron Stars and How Femtoscopy Can Address It* (May 2026)

Presented *Searching for BSM Supersymmetry Using Multi-Lepton Jet Events at the CMS Experiment* (May 2026)

Presented *Evidence for Direct CP Violation in Baryons at the LHCb Experiment* (Apr. 2026)

Outreach

Advising Graduate/Undergraduate Researchers on National and International Opportunities

Alumni Panelist,

U.S. Dept. of Energy Office of Science Graduate Student Research Program – International Research Collaboration Allowance Program (DOE SCGSR IRCA) to CERN (Jun. 2026)

University of Houston Summer Undergraduate Research Fellowship (SURF) Program (Jun. 2026)

University of Houston SURF Program (Jun. 2025)

University of Houston SURF Program (Jun. 2023)

Justin Chan,

U.S. Dept. of Energy Office of Science Graduate Student Research Program (DOE SCGSR) 2025 Solicitation 2, Awardee (Mar. 2026)

Christian Schmidt,

University of Houston Summer Undergraduate Research Fellowship (SURF) Program, Awardee (Jun. 2025)

University of Houston Provost's Undergraduate Research Scholarship (PURS), Awardee (May 2026)

U.S. Goldwater Scholarship, Nominee (Feb. 2026)

Nuclear Physics D.C. Day, Washington D.C., Apr. 2025

Nuclear Physics Research Advocate,

Advocated for the FY2025 U.S. Department of Energy Office of Science, U.S. Nuclear Physics Program, and National Science Foundation nuclear physics budgets in meetings with congressional staff from the offices of Sen. John Cornyn (TX), Sen. Ted Cruz (TX), Rep. Lloyd Doggett (TX), and Sen. Todd Young (IN).